



Empirical calibration of earthquake magnitudes for the Tell Atlas (Northern Algeria)

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Abstract

The Tell Atlas of Algeria is an earthquake-prone area, which experienced many strong earthquakes induced by the collision between the African and Eurasian plates. The “Centre de Recherche en Astronomie Astrophysique et Géophysique” (CRAAG), in charge of the Algerian Seismological Survey, provides, for each earthquake, the parameters of the macroseismic intensity (I_0), the epicenter location and the duration magnitude (M_d). The catalog of Algerian earthquakes contains data represented mainly by maximum intensity I_0 (for historical seismicity and mostly pre-instrumental earthquakes era) and duration magnitude M_d for most of the instrumental seismicity. A previous study by Hamdache et al. (Hamdache et al., Seismol Res Lett 81:732–739, 2010) produced a unified catalog of the main seismic event in Algeria. This study aims to homogenize the Algerian seismic catalog and to achieve its completeness by determining empirical relations converting M_d and I_0 (given by CRAAG) to M_w , M_s , m_b (given by international seismological agencies). To compute the relationship between intensity and magnitudes, we adopted Linear Orthogonal Regression applied to the dataset we selected. Our catalog is now complete and homogeneous, and M_w , M_s and m_b were assigned for historical earthquakes. This will be useful for earthquake engineering and for those dealing with seismic hazard assessment in the Tell Atlas (Algeria), where several urbanized large cities are under the threat of strong earthquake, as is the case for Algiers capital city. The recent urban development in countries with catalogs limited primarily to macroseismic data increased the need for useable relations to convert epicentral intensities to instrumental magnitude m_b , M_s or M_w .

Keywords Algeria · Intensity–magnitude · Linear regression · Orthogonal regression · Earthquake catalog

1 Introduction

The paper investigates the derivation of formulas converting duration magnitude and macroseismic intensity reported by the Algerian catalog (for Algerian earthquakes) into M_w (moment magnitude), M_s (Surface-wave magnitude) and m_b (body-wave magnitude) for the seismicity that occurred in the Tell Atlas (northern Algeria). Algeria’s seismological catalog is composed mainly of epicentral macroseismic intensity (I_0) for historical and most old instrumental

earthquakes and the duration magnitude (M_d) for the instrumental-era seismicity. To homogenize and complete parameters of the Algerian seismological catalog, we computed empirical relations to convert M_d and I_0 into M_w , M_s , and m_b . We computed these relationships by adopting linear orthogonal regression applied to selected events from the Algerian catalog. The seismic parameters I_0 and M_d reported by the seismological bulletins of the CRAAG were, thus, converted to the corresponding magnitudes calculated by two selected international seismological agencies, the *International Seismological Center* (ISC-IDC) and the *Eidgenössische Technische Hochschule Zürich* (ETHZ, Switzerland) (see Data and Resources). The results obtained and the reliability of the formulas are discussed. This led us to update our database with respect to M_w and complete it with the other corresponding magnitudes M_s , m_b , especially for historical earthquakes. In countries with catalogs limited primarily to macroseismic data, the need for useable relations to convert epicentral intensities to instrumental magnitude m_b , M_s or

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M_w is high because earthquake physics is better accounted for by a single parameter which is magnitude. This is the reason why in this study, we convert the epicentral intensities into magnitude so that they could be used for earthquake engineering purposes and seismic hazard assessment studies in the Tell Atlas (Algeria), where several urbanized large cities are under the threat of strong earthquake as for Algiers capital city and other urban cities, which experienced strong and damaging earthquakes in the past.

1.1 Seismicity of the Tell Atlas of Algeria

The seismicity of Algeria is concentrated along the Africa–Eurasia plate boundary collision zone (Fig. 1), as reported by different studies (Benhallou 1985; Ayadi and Bezzeghoud 2015; Harbi et al. 2015). This shallow seismicity is characterized by moderate to strong earthquakes and is concentrated along the Tell Atlas (northern Algeria), extending from the Algerian coastal margin to the South Atlantic faults system. This area is a site of a significant crustal deformation characterized by predominant reverse faulting in the western and central parts and strike-slip faulting in the eastern part. The study area experienced several strong and destructive earthquakes striking different cities, such as those of Algiers (1365, $I_0 = X$ MM; Ayadi and Bezzeghoud 2015), Djidjelli (1856,

$I_0 = IX$ EMS; Ambraseys 1982), M'Sila (1885, $I_0 = IX$ MSK; Harbi et al. 2010), Sidi Aich (1901, $I_0 = IX$ MM; Harbi et al. 2010), El Asnam (1980, $M_s = 7.3$; Yielding et al. 1989), Constantine (1985, $M_s = 5.9$; Ousadou et al. 2013), Mascara (1994, $M_s = 6.0$; Ayadi et al. 2002) and Zemmouri (2003, $M_w = 6.8$; Ayadi et al. 2008). The Algerian seismic catalog displays both historical and instrumental data. Due to the sparse distribution of seismographs over the area of study, the size of most of the seismic events prior to 1990 are solely given with epicentral macroseismic intensity, which represents the unique estimator of energy released during the earthquake.

At present, the estimation of the magnitude of the Algerian earthquakes is given using the duration of the seismic signal. This duration magnitude $M_{d(CRAAG)}$, reported by the Algerian catalog, related to the duration of the seismic signal, is the only magnitude calculated by the CRAAG seismological service using the relation below modified for Algeria from the formula proposed by Lee et al. (1972).

$$M_{d(CRAAG)} = -1.90 + 2.8 \log(\tau) + 0.0035 (\Delta),$$

where (τ) is the total duration of the seismic signal in seconds and (Δ) represents the epicentral distance in km. This relation allows rapid magnitude estimation only for local events ($\Delta \leq 100$ km). The other type of magnitudes (M_s , m_b ,

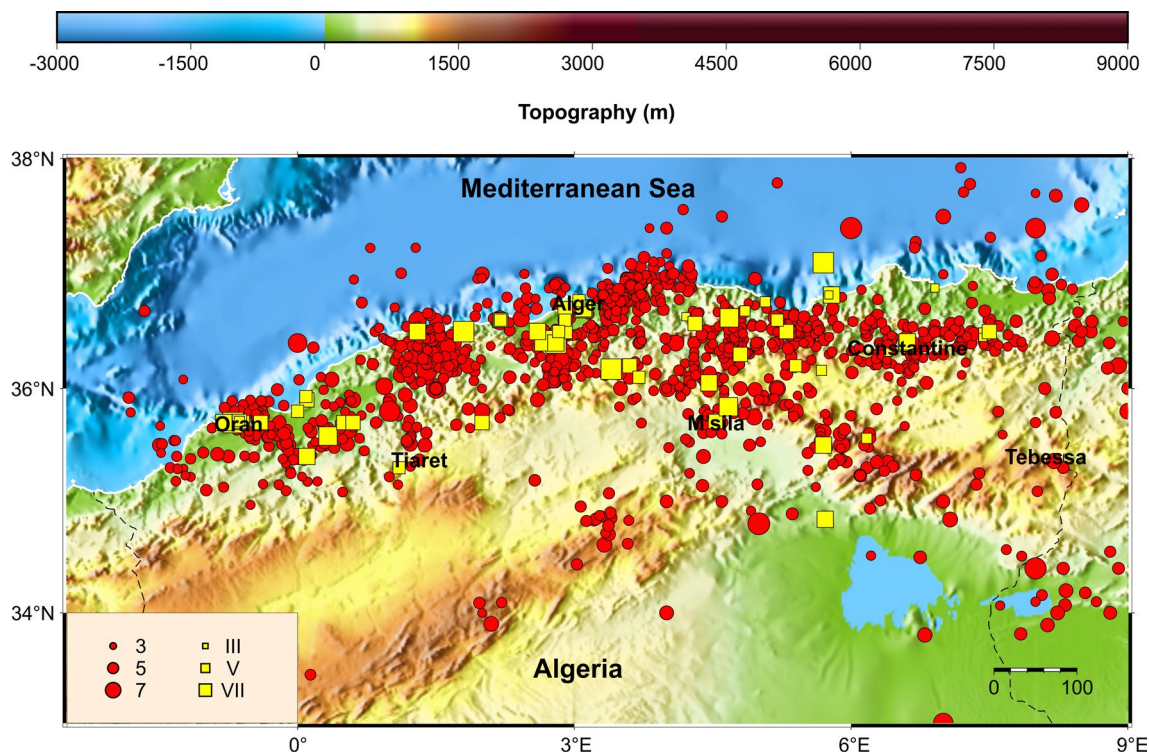


Fig. 1 Seismicity of northern Algeria between 1365 and 2020 from CRAAG database. Yellow squares for the seismicity prior to 1990 and red circles for the seismicity after 1990

Table 1 Summary of empirical relations developed by other studies for Algeria

Authors	Empirical relations	Validity range	Number of events	Average error
Mokrane et al. (1994)	$M_m = 0.51 I_0 + 1.25$ $M_m = 0.97 I_0 - 2.24$	$I_0 \leq \text{VIII}$ $\text{IX} \leq I_0 \leq \text{XI}$	36	
Benouar (1994)	$M_s = 1.18 I_0 + 1.36$		262	± 0.50
Boughacha (2005)	$I_0 = 1.95 m_b - 2.74$		131	

M_w) contained in the Algerian catalog are recovered from international seismological agencies' catalogs.

Since the introduction of the M_w (moment magnitude) based on the seismic moment and energy concept (Kanamori and Brodsky 2004), this parameter has become the most used to quantify the magnitude of an earthquake avoiding the saturation of the Richter magnitude scale, especially for high values of magnitude. The earthquake magnitude is directly related to the seismic moment M_0 . Here, we determine the laws scaling both M_d and I_0 (for the Algerian events) with corresponding M_s , m_b and M_w given by international agencies (ISC and ETHZ) (see Data and Resources).

1.2 Algerian earthquake catalog

The Algerian earthquake catalog covers the period AD125 to the present and provides standard parameters such as date, time of occurrence, hypocenter coordinates (latitude, longitude and depth), the maximum intensity I_0 at epicenter and often completed by the magnitude M_d of the earthquake. The macroseismic intensity I_0 at epicenter is defined as a classification of the strength of the earthquake shaking in terms of related observed effects and damage. This parameter in our catalog is expressed using different scales. The most used are: Modified Mercalli scale "MM", Medvedev–Sponheuer–Karnik scale "MSK" and the recent European Macroseismic Scale 1998 "EMS98". Another intensity scale called "Rossi-Forel" was also used in Algeria at the beginning of the 1900s. In this work, all the intensities contained in the Algerian catalogs were converted to EMS98 scale (Harbi et al. 2015). It is noteworthy that before implementing the first seismic stations (in 1910) at Bouzaréah (Algiers), the seismological studies were confined into macroseismic studies. Although the number of seismological stations increased irregularly between 1910 and 1990, the study of the seismic event continued to be assessed following the macroseismic approach. Thus, as for many other countries, the most used parameter in the Algerian seismic catalog for both historical and recent earthquakes was the macroseismic intensity I_0 at epicenter, based on the description of effects and damage caused by the earthquake. Recently, past events were reappraised on the basis of original sources such as unpublished manuscripts, chronicles, newspapers, and periodicals for epicentral intensity I_0 re-estimation (Harbi et al.

2015). Even if the instrumental era began with the first seismological station installed in 1910 (Bouzaréah–Algiers), the national seismograph network of Algeria was upgraded first in 1990 with 32 telemetered seismological stations (Bezzeghoud et al. 1994). Since that date, duration magnitude M_d computed at CRAAG was assigned for each seismic event using the duration of its recorded signal. In this study, we consider the available instrumental data for about 30 years (1981–2014) to compute a formula scaling duration magnitude M_d and I_0 with other magnitudes calculated by international seismological agencies (see Data and Resources).

Since the beginning of the twentieth century, many seismographs were introduced to record the amplitude of ground motion, and different magnitude scales were proposed; M_L , m_b , M_s , M_d . The inconvenience of these magnitude scales is the saturation for a certain threshold related to strong earthquakes. To overcome this limitation, a non-saturating magnitude scale M_w was proposed by Kanamori (1977) based on the total scalar seismic moment M_0 responsible for the rupture during an earthquake and calculated using the formula below:

$$M_w = 2/3 (\log M_0 - 10.7),$$

while the seismic moment M_0 is defined by

$$M_0 = DS$$

where μ is the rigidity; D is the average offset on the fault; and S is the area of the fault plane.

The update of our catalog with M_w will be useful, especially for earthquake engineering studies.

1.3 Previous studies on magnitude scaling in Algeria

To mitigate the problem of homogenization and parameter completeness of Algerian seismic catalog, several studies were conducted to establish empirical laws scaling the different parameters I_0 , M_d , M_s and m_b .

Mokrane et al. (1994) established a linear relationship between the magnitude m_b and I_0 and named it the macroseismic magnitude M_m (Table 1). This relation transforms epicentral intensity into magnitude considering two ranges of intensities (Table 1).

Benouar (1994) compiled a seismic catalog for the Maghreb area from 1900 to 1990. This catalog led the author to calibrate M_s to I_0 , as given in Table 1.

More recently, Boughacha (2005) proposed a formula scaling I_0 to m_b using a collection of data from the catalogs of Mezcua and Martinez (1983), Benhallou (1985) and Mokrane et al. (1994). Unfortunately, Mokrane et al. (1994) and Boughacha (2005) did not give any indication about error and correlation coefficient (Table 1).

In this study, we proposed a new scaling formula for M_d and I_0 with respect to M_w , M_s , m_b to complete the Algerian seismic catalog.

2 Database

The analysis presented in this study is based on the seismicity data obtained from the compilation of earthquakes parameters available from several references and sources: Ayadi and Bezzeghoud (2015), Bezzeghoud et al. (2014), Boughacha (2005), Braunmiller and Bernardi (2005), Centre de Recherche en Astronomie, Astrophysique et Géophysique (CRAAG), Eidgenössische Technische Hochschule Zürich (ETHZ), European Mediterranean Seismological Centre (CSEM), Harbi et al. (2010), Harbi et al. (2015), Instituto Geográfico Nacional (IGN), International Seismological Centre (ISC) and Stich et al. (2003) (see Data and Resources).

The database we used contains 115 events that occurred between 1954 and 2020 in the area of study (2°W – 9°E , 33° – 38°N). This database provides the following information: *date and time* of seismic events; the hypocenter location and the focal depth h ; the macroseismic intensity I_0 , which varies between IV and X (EMS98 scale); instrumental magnitudes (M_d , M_w , M_s and m_b) with the reference reporting the event. Most of these earthquakes is characterized by moderate magnitude ($M < 6$) with few strong seismic events. The highest value of duration magnitude M_d (CRAAG) is M_d 6.4 attributed to the Zemmouri earthquake 2003, and M_s 7.3 for the El Asnam earthquake 1980. All the events in our database are shallow depths ($h < 30$ km).

3 Methodology

The methodology adopted in this study was inspired by the work of the late Peter Bormann (Bormann 2015), which consists of the selection of data pairs formed by data extracted from the CRAAG catalog (I_0 and M_d) and their corresponding magnitudes (M_s , m_b and M_w) calculated by the selected seismological agency (see Data and

Resources). However, to avoid introducing errors due to the multiplicity of sources, the first step was to filter the dataset in order to keep only the data pairs from CRAAG and the corresponding agency for a specific type of magnitude. We plot the selected data associated with the pairs [M_d (CRAAG); selected magnitude (selected agency)] or [I_0 (CRAAG); selected magnitude(selected agency)]. These data pairs are used to examine the linear relationship between the considered parameters. The linear regression used is given by the equation below:

$$Y = aX + b. \quad (1)$$

Instead of using the common least squares linear regression, considering the fact that the dependent variable (y) and independent variable (x) are affected by uncertainties, we preferred to use the linear orthogonal regression (Castellaro et al. 2006; Castellaro and Bormann 2007; Carr 2012; Lolli and Gasperini 2012). In Eq. 1, X and Y are the true values of the considered parameters (magnitude or intensity)

$$\text{with } y = Y + \varepsilon_y, \quad (2)$$

$$\text{and } x = X + \varepsilon_x, \quad (3)$$

y and x are the measured or observed values, and ε_y and ε_x are the errors that affect Y and X , respectively.

The method minimizes the following sum:

$$\sum_{i=1}^n \{(y_i - b - aX_i)^2 / \rho + (x_i - X_i)^2\}, \quad (4)$$

$$\text{with } \rho = \frac{\varepsilon_x^2}{\varepsilon_y^2}. \quad (5)$$

The difference between the standard least squares regression and orthogonal regression is that orthogonal regression minimizes the sum of squared perpendicular distances from the data to the regression line. Considering the perpendicular distance means that we take into account both data errors on both y and x axes. Meanwhile, the least squares regression minimizes the sum of squares distances between the data point and the regression line but is parallel to the y axis only.

The linear orthogonal regression we applied to our data set gave us a good correlation coefficient (R) and standard deviation σ for each empirical law.

The estimate of the coefficients of Eq. (1) a , b , R and standard deviation indicate the regression goodness of the dataset. We have applied this procedure to derive meaningful and reliable conversion relationships between the parameters of earthquakes (M_d , I_0 , m_b , M_s and M_w) from the Algerian seismic catalog data.

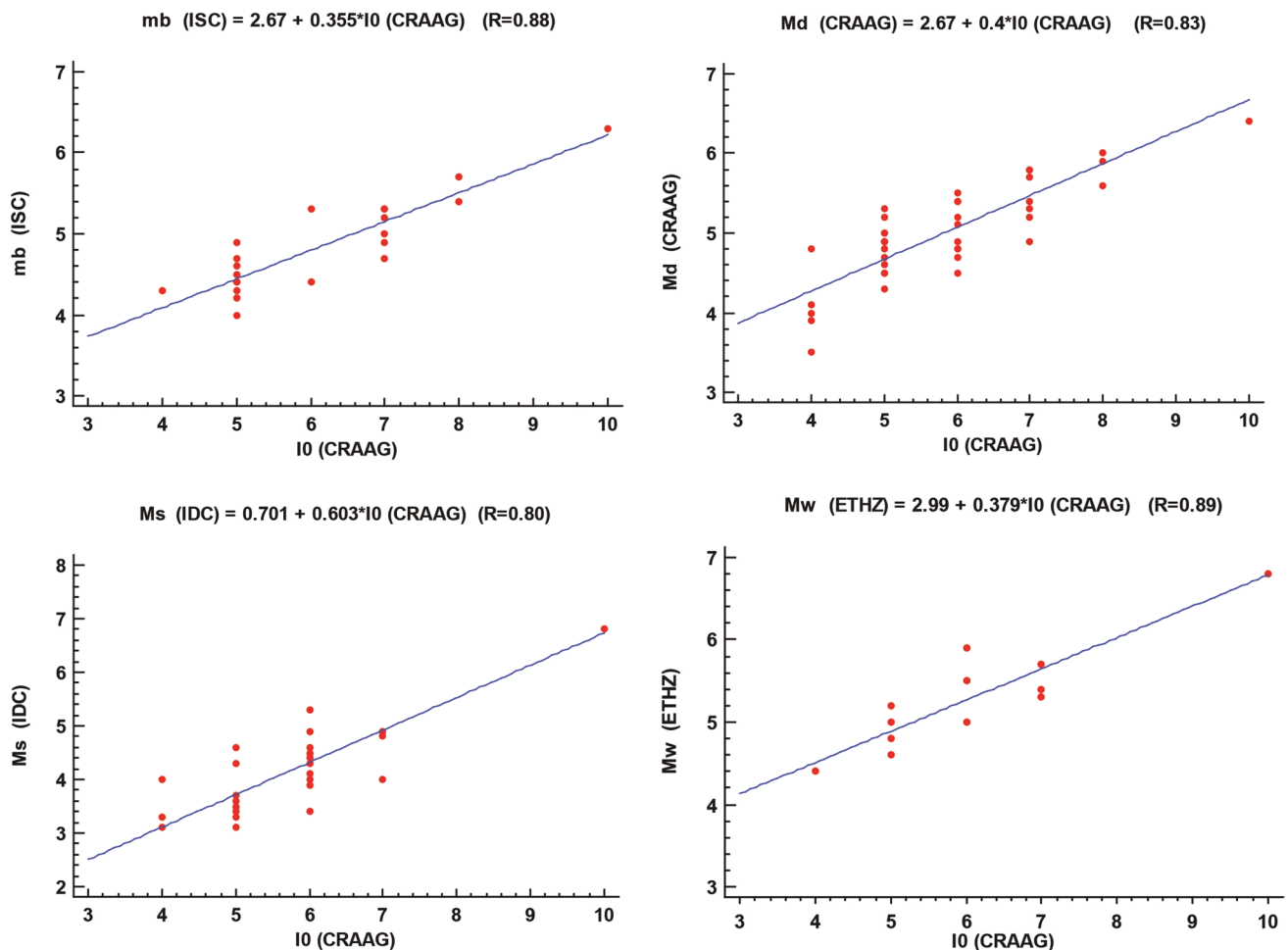


Fig. 2 Empirical laws scaling I_0 with m_b , M_s , M_w and M_d for Algeria

4 Results

Following the adopted methodology, we applied the orthogonal regression to our database, and obtained a good approximation, as shown in Figs. 2 and 3.

For epicentral intensity and magnitude (M_d , M_s , m_b , M_w), the linear approximation is given with a correlation coefficient ranging between $0.81 \leq R \leq 0.89$ (Table 2).

For scaling the epicentral intensity with the duration magnitude, we calculated the relation below:

$$M_d(CRAAG) = 0.40I_0(CRAAG) + 2.67(\pm 0.33). \quad (6)$$

This linear regression illustrated in Fig. 2 is obtained with a correlation coefficient of $R=0.83$ and $\sigma=0.33$ units. The approximation of the dataset by Eq. 6 shows a good shape with dots less scattered with respect to the curve. This is the best approximation proposed for scaling epicenter intensity with duration magnitude calculated by CRAAG. We have no comparison with other relations for Algeria since it is

the first time such relations are established. The intensity I_0 calculated at CRAAG was also converted to m_b taken from ISC database to the 21 events found. The linear regression as given in Fig. 2 shows a good correlation coefficient of $R=0.88$ and $\sigma=0.27$. The regression law here is illustrated by the following relation:

$$m_b(ISC) = 0.36I_0(CRAAG) + 2.67(\pm 0.27). \quad (7)$$

The epicentral intensity estimated by CRAAG was also compared to moment magnitude following the same procedure, and we have determined the relation scaling M_w with I_0 represented in Fig. 2. The relation (8) is derived by a combination of 12 seismic events for 1999–2010, with $R=0.89$ and $\sigma=0.29$ units. The M_w values were taken from the ETHZ bulletin.

$$M_w(ETHZ) = 0.38I_0(CRAAG) + 2.99(\pm 0.29). \quad (8)$$

The CRAAG epicenter intensity (I_0) was also scaled with M_s by the relationship (9) represented in Fig. 2 on the

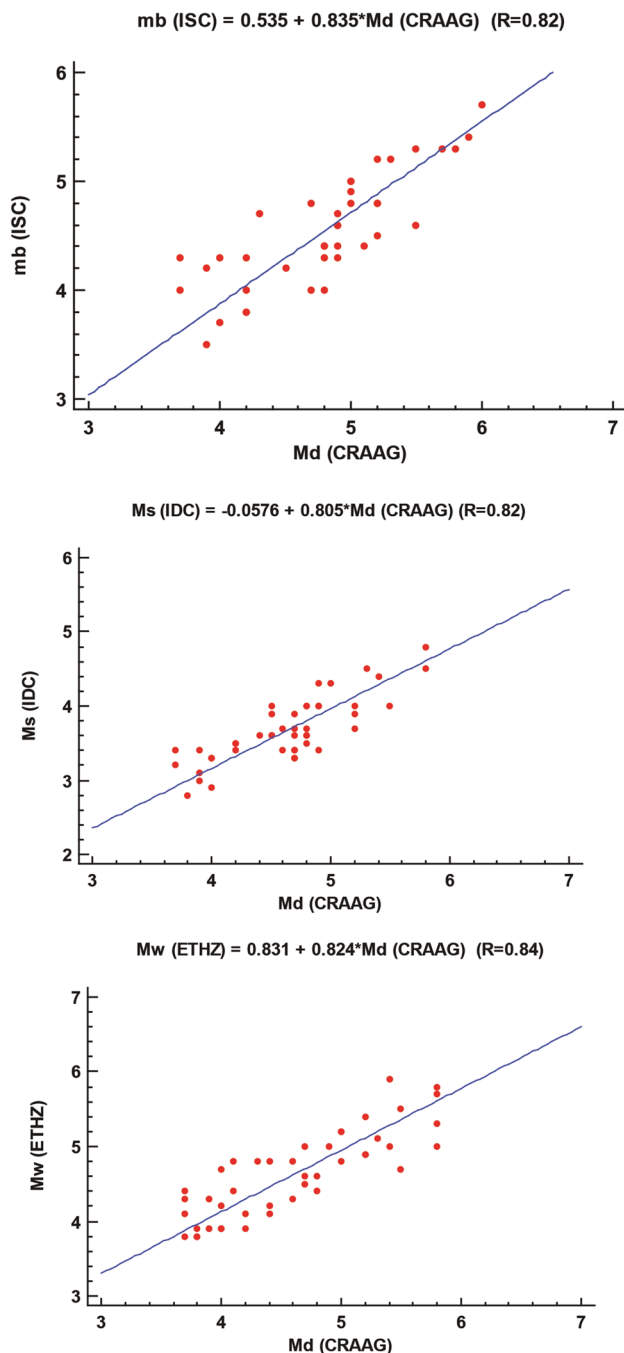


Fig. 3 Conversion duration magnitude M_d with other magnitudes m_b , M_s and M_w for Algeria

basis of 24 earthquakes that occurred between 2000 and 2014, with $R=0.80$ and $\sigma=0.50$, while M_s magnitudes are extracted from ISC (IDC) catalog.

$$M_s(\text{ISC}) = 0.60I_0(\text{CRAAG}) + 0.70(\pm 0.50). \quad (9)$$

Despite the few earthquakes used, the relationships fit better the data pairs (Fig. 2). As we can notice, the correlation

Table 2 Summary of the relationship between I_0 and M_d from CRAAG database with m_b , M_s and M_w from selected sources

Empirical relations for Algeria	References	Period	R	σ
$M_d=0.38 I_0+2.80$	CRAAG	1981–2014	0.83	0.34
$M_w=0.37 I_0+3.06$	ETHZ	1999–2010	0.89	0.30
$M_s=0.53 I_0+1.11$	IDC	2000–2014	0.81	0.50
$m_b=0.34 I_0+2.73$	ISC	1967–2014	0.88	0.27
$M_w=0.71 M_d+1.34$	ETHZ	1999–2010	0.84	0.31
$M_s=0.68 M_d+0.50$	IDC	2000–2014	0.82	0.26
$m_b=0.71 M_d+1.14$	ISC	1985–2012	0.82	0.30

coefficient values greater than 0.80 indicate the non-dispersion between the straight lines obtained and the dataset.

The duration magnitude M_d calculated at CRAAG is also converted to another type of magnitude. A conversion relation to M_w was determined. The moment magnitude M_w in this study is from the ETHZ. Figure 3 shows the relationship between M_w and M_d (Eq. 10) with a set of data for 1999 to 2010 using 44 samples with magnitudes ranging from 3.7 to 5.9. This compilation yields the following formula with $R=0.84$ and $\sigma=0.32$.

$$M_w(\text{ETHZ}) = 0.82M_d(\text{CRAAG}) + 0.83(\pm 0.32). \quad (10)$$

Duration magnitude M_d was scaled with M_s by (11) as illustrated in Fig. 3. The M_s were extracted from ISC (IDC) bulletins. This relation (Eq. 11) is derived using 45 seismic events in the magnitude range 2.8 to 5.8 that occurred during the period 2000–2014, with $R=0.82$ and $\sigma=0.26$.

$$M_s(\text{ISC}) = 0.80M_d(\text{CRAAG}) - 0.06(\pm 0.26). \quad (11)$$

The duration magnitude and body-waves magnitude could also be linked by regression as given by Eq. 12. A selection of 39 earthquakes that occurred between 1985 and 2012 with magnitude ranging between 3.5 and 6.0. The equation 12 that convert M_d into m_b is given in Fig. 3, with $R=0.82$ and $\sigma=0.30$, with m_b data reported from ISC bulletin.

$$m_b(\text{ISC}) = 0.83M_d(\text{CRAAG}) + 0.53(\pm 0.30). \quad (12)$$

Figure 3 shows the goodness of the fits of the datasets with R greater than 0.80 and σ does not exceed 0.31, which means that Eq. 12 correlates very well with observations.

All the plots show a good distribution of data points around the regression line strengthening, the approximation obtained.

All these relations scaling epicenter intensity (I_0), duration magnitude (M_d), surface-waves magnitude (M_s), and moment magnitude (M_w) enabled us to complete the Algerian seismic catalog and give a tool to those working on seismic hazard assessment to better evaluate their model for earthquake engineering purposes.

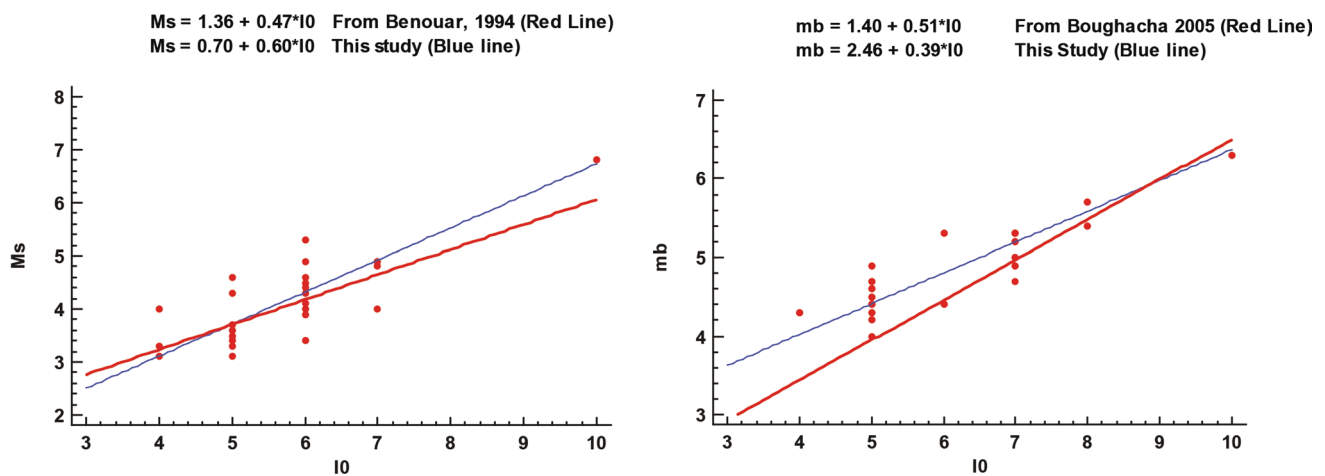


Fig. 4 Comparison between: I_0 versus M_s given by Benouar (1994) and I_0 versus m_b given by Boughacha (2005) with equivalent relations established in this study

5 Conclusion

The majority of the seismic events that occurred in northern Algeria during the historical period have only epicentral macroseismic intensity. For seismic hazard assessment and earthquake engineering studies that need an earthquake catalog with moment magnitude, body or surface wave magnitude (M_w , M_s and m_b), a conversion of the macroseismic parameters is necessary.

For a long time, the Algerian catalog provided duration magnitude M_d and macroseismic intensity I_0 given by CRAAG. This led us to look for empirical relationships scaling these parameters with magnitudes. To avoid biases introduced by data provided by multiple sources (Agencies), we derived our relations with respect to the magnitude reported by a single agency (M_s given by ISC and m_b given by IDC and the moment magnitude M_w calculated by ETHZ). The relations obtained will be used as empirical scaling laws to homogenize the Algerian earthquake catalog summarized in Table 2. The calibration of magnitudes for historical earthquakes is of great importance in seismic hazard assessment studies by the quantification of earthquakes size that occurred before the instrumentation era. The empirical relations obtained in this study are reliable regarding the goodness of the fit of the regression lines with respect to the data points. Comparing with the relations obtained in previous studies the equations obtained in this study are more reliable, choosing a parameter from a single agency reduces the biases that could be related to procedures, operators, instrument response and epicenter–instrument distance, generally for the same event we have many values for M_s , m_b and M_w given by several agencies.

Comparing our relations with that proposed by Benouar (1994) for I_0 – M_s conversion and Boughacha et al. (2004) for

I_0 – m_b conversion (Fig. 4), we can see that our results show significant differences due to the multiplicity of sources and the intensity scales used by the authors.

The linearity of I_0 versus all magnitudes is evidenced by the graphs and supported by the relations obtained with an estimate of the standard deviation (Table 2).

With our relations scaling the epicentral intensity and duration magnitude (calculated at CRAAG) with the magnitudes M_s , m_b and M_w (calculated by ISC, IDC and ETHZ), the update of the Algerian seismic catalog is now possible.

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Data and resources The data used in this study were obtained from the Algerian seismic catalog (Mokrane et al. 2004). The data are also available from the electronic supplement of Ayadi and Bezzeghoud (2015) (www.seismosoc.org/Publications/SRL/SRL_86/srl_86-ayadi_and_bezzeghoud-esupp/) and Harbi et al. (2015) (www.seismosoc.org/Publications/SRL/SRL_86/srl_86-6-harbi_et_al-esupp/). Other published papers were also used in this study such Ayadi and Bezzeghoud (2015), Harbi et al. (2015), Mokrane et al. (1994), Benouar (1994), Benhalou (1985). We have also obtained data from: Centre de Recherche en Astronomie, Astrophysique et Géophysique. Euro-Mediterranean Seismological Centre (www.emsc-csem.org). Eidgenössische Technische Hochschule Zürich (<http://seismo.ethz.ch/en/home/>). International Data Centre (IDC), International Seismological Centre (<http://www.isc.ac.uk/>). National Earthquake Information Center (<https://earthquake.usgs.gov/earthquakes/search/>). Figure 1 was made using the Generic Mapping Tools version 4.2.1 (www.soest.hawaii.edu/gmt/; Wessel and Smith 1998). The other figures were drawn using MATLAB.

Data availability statements The data used in this study are available upon request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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